

A project report on

MULTI-DISEASE CHEST X-RAY CLASSIFICATION USING DEEP LEARNING AND EXPLAINABLE AI

Submitted in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

by

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April, 2026

DECLARATION

I hereby declare that the thesis entitled “**MULTI-DISEASE CHEST X-RAY CLASSIFICATION USING DEEP LEARNING AND EXPLAINABLE AI**” submitted by me, for the award of the degree of **BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING** VIT is a record of bonafide work carried out by me under the supervision of Dr. Yelepi Usha Rani.

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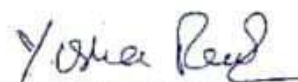
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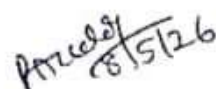
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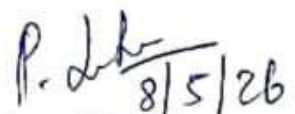
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ABSTRACT

Chest X-ray imaging is widely used for diagnosing lung and chest-related diseases such as pneumonia, cardiomegaly, and pleural effusion. However, manual interpretation can be time-consuming and challenging, especially when multiple diseases are present in a single image. This highlights the need for automated systems to support medical professionals.

With advancements in Artificial Intelligence, particularly deep learning, automated medical image analysis has improved significantly. Convolutional Neural Networks (CNNs) are effective in extracting features from images and performing classification tasks. In this project, a multi-label chest X-ray classification system is developed using transfer learning with pre-trained models such as DenseNet121, EfficientNet-B0, MobileNet, and VGGNet.

The system is designed to detect multiple diseases from a single image using a sigmoid activation function, which provides independent probability scores for each class. Binary Cross-Entropy loss is used to handle the multi-label classification problem effectively. The dataset is preprocessed through resizing, normalization, and augmentation techniques to improve model performance and generalization.

To enhance interpretability, Grad-CAM is applied to generate heatmaps that highlight important regions in the X-ray images influencing the model's predictions. Experimental results show that pre-trained models perform well in multi-disease detection, with MobileNet achieving strong accuracy while maintaining computational efficiency. The system is also capable of filtering invalid inputs to ensure only relevant X-ray images are processed. This improves reliability and reduces incorrect predictions. The approach can be integrated into clinical workflows to assist doctors in decision-making. Additionally, the system supports faster screening of large volumes of X-ray data. It reduces the burden on radiologists by providing preliminary automated analysis. The model can also be extended to include more disease classes in future work.

Overall, the proposed system provides a reliable and efficient approach for chest X-ray disease classification and can assist healthcare professionals in faster and more accurate diagnosis.

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LIST OF ACRONYMS

AI	Artificial Intelligence
CNN	Convolutional Neural Network
DL	Deep Learning
X-ray	X-ray Imaging Technique
NIH	National Institutes of Health
GAP	Global Average Pooling
ReLU	Rectified Linear Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
BCE	Binary Cross Entropy
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
PR	Precision-Recall
TPR	True Positive Rate
FPR	False Positive Rate
GPU	Graphics Processing Unit

CHAPTER 1

INTRODUCTION

Medical imaging has become an essential part of modern healthcare, helping doctors diagnose diseases quickly and accurately. Among different imaging techniques, chest X-rays are one of the most widely used methods for examining lung and heart conditions. They are commonly used because they are affordable, fast, and available in almost all hospitals.

Even though chest X-rays are useful, analyzing them is not always easy. The images often contain overlapping structures such as ribs, tissues, and organs, which can make abnormalities difficult to identify. In addition, the visual differences between normal and diseased conditions can be very small, requiring experienced radiologists for accurate interpretation.

Another important issue is the increasing number of patients and limited availability of expert radiologists. Doctors are often required to examine many X-ray images in a short time, which can lead to delays and inconsistencies in diagnosis. In some cases, early signs of diseases may be missed due to workload or fatigue.

A key challenge in chest X-ray analysis is that a single image can show multiple diseases at the same time. For example, a patient may have both pneumonia and pleural effusion. However, many traditional systems and earlier machine learning models are designed to detect only one disease per image. This limitation reduces their effectiveness in real-world medical scenarios.

Recent advancements in Artificial Intelligence, especially Deep Learning, have opened new possibilities in medical image analysis. Convolutional Neural Networks (CNNs) have shown strong performance in image classification tasks by automatically learning patterns and features from images. These models can help in identifying diseases with good accuracy and speed.

Instead of building deep learning models from scratch, many modern approaches use **transfer learning**, where pre-trained models are adapted for a specific task. These models have already learned useful image features from large datasets and can be fine-tuned for

medical applications. This approach reduces training time and improves performance, especially when the available dataset is limited.

In this project, a system is developed to analyze chest X-ray images using **pre-trained CNN models**. The focus is on **multi-label classification**, where the model can detect multiple diseases from a single image. This makes the system more practical and closer to real-world clinical conditions.

To make the model more reliable and understandable, an explainable AI technique called **Grad-CAM** is used. This method highlights the important regions in the X-ray image that influence the model's prediction, helping users understand how the decision is made.

1.1 OBJECTIVES

The main objectives of this project are:

- To develop a system that can detect **multiple diseases from a single chest X-ray image** using multi-label classification
- To apply **transfer learning with pre-trained CNN models** for effective medical image analysis
- To train and compare different pre-trained models such as DenseNet, EfficientNet, and MobileNet
- To preprocess chest X-ray images to improve model performance and consistency
- To evaluate the models using **accuracy and loss values**
- To incorporate **Grad-CAM visualization** to understand and explain model predictions

1.2 BACKGROUND AND LITERATURE SURVEY

Chest X-ray analysis has been an active area of research in medical image processing due to its importance in diagnosing various thoracic diseases. In recent years, deep learning techniques, especially Convolutional Neural Networks (CNNs), have significantly improved the performance of automated disease detection systems.

Early approaches relied on traditional machine learning methods, which required manual feature extraction. These methods were limited in performance because they could not

capture complex patterns present in medical images. With the introduction of deep learning, models are now capable of automatically learning important features directly from raw images, leading to better accuracy and reliability.

A study by **A. S. Pillai (2022)** proposed a deep learning-based system for multi-label chest X-ray classification, where a single image can be associated with multiple diseases. The work focused on predicting up to 14 different diseases using CNN models and highlighted the importance of multi-label classification in real-world clinical scenarios. The study also emphasized that large datasets such as NIH ChestX-ray14 play a key role in improving model performance.

Another important contribution by **Jakub Kufel et al. (2023)** used EfficientNet with transfer learning for multi-label classification. Their work addressed the issue of dataset imbalance by using a custom data split and achieved strong performance using AUC-ROC metrics. The study demonstrated that transfer learning significantly improves model efficiency and reduces computational requirements while maintaining high accuracy.

Research by **Yufei Jin and Huijuan Lu (2023)** explored deep learning-based multi-disease classification using CNN architectures. Their findings showed that deep learning models perform better when trained on large datasets and are capable of handling multiple disease predictions effectively. They also highlighted that multi-label classification better represents real-world medical conditions.

Another study by **Baltruschat et al. (2019)** compared different deep learning architectures for chest X-ray classification. The results indicated that transfer learning improves performance, especially when training data is limited. The study provided a strong baseline for comparing different CNN models.

In addition, **Hongyu Wang et al. (2020)** introduced a triple attention mechanism to improve the focus of deep learning models on relevant regions of the image. This approach enhanced the interpretability and accuracy of disease detection by allowing the model to concentrate on important features.

Recent advancements also include transformer-based models such as **SwinCheX (Taslimi, 2022)**, which capture global relationships in images. These models have shown promising results but require high computational resources compared to CNN-based approaches.

From the above literature, several key observations can be made:

- Deep learning models outperform traditional methods in chest X-ray analysis
- Multi-label classification is necessary for handling real-world medical cases
- Transfer learning improves model performance and reduces training time
- Dataset imbalance is a major challenge affecting prediction accuracy
- Explainability is important for clinical acceptance of AI models

Despite these advancements, many existing systems either focus on single-disease classification or lack proper interpretability. Therefore, this project focuses on using **pre-trained CNN models with transfer learning and Grad-CAM visualization** to develop an efficient and explainable multi-label classification system.

1.3 ORGANIZATION OF THE REPORT

The remaining chapters of this project report are organized as follows:

- **Chapter 2** presents the proposed system, overall methodology, and software tools used in the project.
- **Chapter 3** describes the implementation details, including model training and system development.
- **Chapter 4** discusses the results obtained and provides analysis of the model performance.
- **Chapter 5** provides the conclusion of the project along with key observations.
- **Chapter 6** includes appendices containing supplementary information related to the project.
- **Chapter 7** lists the references used in this work in the required format.

CHAPTER 2

PROPOSED SYSTEM AND METHODOLOGY

This chapter describes the proposed system, working methodology, and implementation details of the project.

2.1 PROPOSED SYSTEM

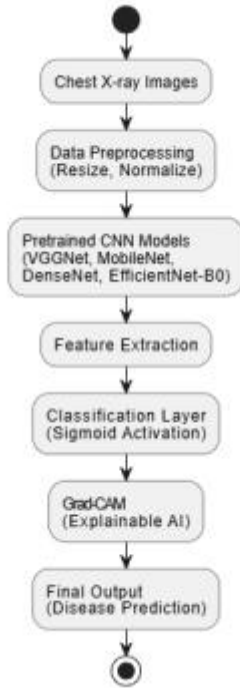


Figure 2.1: Block Diagram

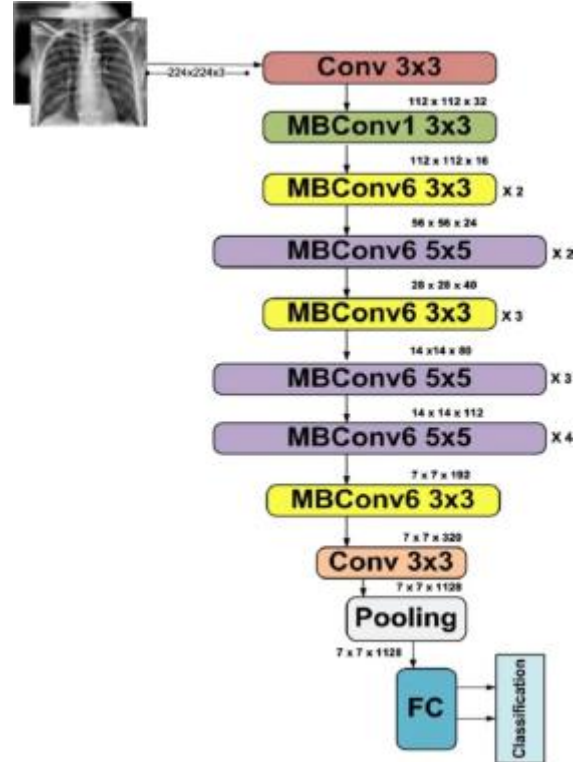


Figure 2.2: EfficientNet-B0 Architecture

The proposed system is a **multi-label chest X-ray disease classification model** developed using pre-trained Convolutional Neural Network (CNN) architectures, with a primary focus on **EfficientNet-B0**. The objective of this model is to automatically learn discriminative visual features from chest X-ray images and accurately classify multiple disease conditions.

Convolutional Neural Networks are highly effective for image analysis tasks as they can capture spatial hierarchies through convolution operations. Unlike traditional approaches that rely on handcrafted features, CNNs learn feature representations directly from input images through multiple layers.

EfficientNet-B0 improves the performance of conventional CNNs by introducing a **compound scaling method**, which uniformly scales network depth, width, and resolution. This results in better accuracy with fewer parameters.

The CNN model consists of the following core components:

1. CONVOLUTIONAL LAYERS

The convolutional layer performs feature extraction by applying learnable filters over the input image. The convolution operation is mathematically defined as:

$$y(i, j) = \sum_m \sum_n x(i + m, j + n) \cdot w(m, n) + b$$

where:

- x is the input image
- w is the convolution kernel
- b is the bias term
- $y(i, j)$ is the output feature map

These layers extract local features such as edges, textures, and patterns.

2. ACTIVATION FUNCTION (RELU)

After convolution, a non-linear activation function is applied to introduce non-linearity into the model. The Rectified Linear Unit (ReLU) is defined as:

$$f(x) = \max(0, x)$$

This allows the network to learn complex relationships within the data.

3. BATCH NORMALIZATION

Batch normalization is used to stabilize and accelerate training by normalizing the inputs of each layer:

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

where:

- μ is the batch mean
- σ^2 is the batch variance
- ϵ is a small constant for numerical stability

4. POOLING LAYERS

Pooling layers reduce the spatial dimensions of feature maps while retaining important information. For max pooling:

$$y = \max(x_{ij})$$

Pooling helps in reducing computational complexity and overfitting.

5. MBCONV BLOCKS (EFFICIENTNET-B0)

EfficientNet-B0 uses **Mobile Inverted Bottleneck Convolution (MBConv)** blocks, which improve efficiency and performance. These blocks include:

- **Depthwise Convolution:**

$$y_{i,j,k} = \sum_{m,n} x_{i+m,j+n,k} \cdot w_{m,n,k}$$

- **Pointwise Convolution (1×1 convolution):**

$$y = x * W_{1 \times 1}$$

- **Residual Connection:**

$$y = F(x) + x$$

These operations help in capturing features efficiently with fewer parameters.

6. GLOBAL AVERAGE POOLING (GAP)

Instead of flattening, Global Average Pooling computes the average of each feature map:

$$y_k = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_{i,j,k}$$

This reduces overfitting and improves generalization.

7. FULLY CONNECTED LAYER

The fully connected layer maps extracted features to class scores:

$$z = Wx + b$$

where W is the weight matrix and b is the bias.

8. OUTPUT LAYER (SIGMOID ACTIVATION)

Since this is a multi-label classification problem, a sigmoid function is applied to each output neuron:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

This produces independent probability values for each disease class.

9. LOSS FUNCTION (BINARY CROSS ENTROPY)

The model is trained using Binary Cross Entropy loss:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

where:

- y_i is the true label
- p_i is the predicted probability

Overall, the CNN architecture, particularly EfficientNet-B0, provides an efficient and scalable solution for multi-label chest X-ray classification. Its optimized structure allows it to capture complex visual patterns while maintaining computational efficiency, making it suitable for real-world medical applications.

2.2 WORKING METHODOLOGY

The proposed system follows a structured deep learning methodology for multi-label chest X-ray disease classification. The methodology is designed to ensure efficient data handling, effective feature extraction, and accurate prediction using pre-trained CNN models. It consists of dataset preparation, preprocessing, transfer learning, model training, evaluation, visualization, and explainability.

2.2.1 DATASET PREPARATION AND DESCRIPTION

The dataset used in this work is obtained from Kaggle, which provides a curated version of the NIH ChestX-ray14 dataset. It contains chest X-ray images labeled with multiple disease conditions, making it suitable for multi-label classification.

In this project, a subset of **10 classes** is considered:

- 1) Atelectasis
- 2) Cardiomegaly
- 3) Effusion
- 4) Infiltration
- 5) Mass
- 6) Nodule
- 7) Pneumonia
- 8) Pneumothorax
- 9) Consolidation
- 10) No Finding

Each image may contain one or more labels, making this a **multi-label classification problem**.

DATASET CHARACTERISTICS

- The dataset consists of 112,120 chest X-ray images
- Images are grayscale medical images with varying resolutions
- High similarity between disease classes makes classification challenging
- Presence of class imbalance (some diseases have fewer samples)
- Labels are not mutually exclusive, allowing multiple diseases per image

DATASET SPLITTING

The dataset is divided into:

- Training and validation set (86,524 images, ~77%) – used for model learning and performance monitoring
- Testing set (25,596 images, ~23%) – used for final evaluation

This ensures that the model is evaluated on unseen data and helps prevent overfitting.

2.2.2 IMAGE PREPROCESSING

Preprocessing is an important step to prepare input data for CNN models and improve training efficiency.

- **Image Resizing:** All images are resized to **224 × 224 pixels**, which matches the input requirement of pre-trained CNN architectures.
- **Normalization:** Pixel values are normalized to the range **[0, 1]**, which stabilizes gradient updates and improves convergence during training.
- **Tensor Conversion:** Images are converted into tensor format to be processed by the PyTorch framework.
- **Batch Processing:** Images are processed in mini-batches to improve computational efficiency and enable parallel processing on GPUs.

2.2.3 TRANSFER LEARNING STRATEGY

Training deep CNN models from scratch requires large datasets and high computational resources. To overcome this limitation, **transfer learning** is employed.

In this approach:

- Pre-trained models (trained on ImageNet) are used
- Learned feature representations are reused
- Final classification layers are modified to match the number of output classes

ADVANTAGES OF TRANSFER LEARNING

- Faster convergence
- Better generalization on limited medical data
- Reduced computational cost
- Improved feature extraction capability

2.2.4 MODEL TRAINING

The selected CNN models (**VGGNet, MobileNet, DenseNet, EfficientNet-B0**) are fine-tuned for the classification task.

During training:

- Input images are passed through the CNN model
- Hierarchical features are extracted automatically
- Predictions are generated for each class
- Errors are minimized using backpropagation

The **Adam optimizer** is used for efficient parameter updates.

TRAINING PARAMETERS

- Learning Rate: 0.001
- Batch Size: 32
- Epochs: 20

The training continues until the model reaches convergence and achieves stable performance on the validation dataset.

2.2.5 MULTI-LABEL PREDICTION STRATEGY

Since chest X-ray images may contain multiple diseases, the system uses **multi-label classification**.

- Each class is predicted independently
- Sigmoid activation is applied (as discussed in Section 2.1)
- Output is a probability vector for all classes

THRESHOLDING

To convert predicted probabilities into final class labels, a custom thresholding strategy is applied.

Initially, the probability of the “No Finding” class is evaluated:

- If the probability of “No Finding” is greater than 0.5, the input X-ray is classified as normal, and no further disease prediction is performed.
- If the probability of “No Finding” is less than or equal to 0.5, the model evaluates the remaining disease classes.

For the other nine disease classes, a lower threshold value of 0.275 is used:

- If any disease class has a probability greater than 0.275, it is considered as a possible detected condition.

This two-step thresholding approach improves the reliability of predictions by first confirming normal cases and then identifying potential abnormalities with higher sensitivity.

2.2.6 MODEL EVALUATION

The performance of the models is evaluated using:

- Training Loss, Validation Loss
- Validation Accuracy

These metrics help analyze learning behavior and model generalization.

OBSERVATIONS

- MobileNet achieves the highest validation accuracy
- EfficientNet provides balanced performance and efficiency
- DenseNet shows comparatively lower performance

2.2.7 PERFORMANCE VISUALIZATION

To better understand model behavior, graphical plots are used:

- Accuracy vs Epoch curve
- Loss vs Epoch curve
- Model comparison graph

These plots help in:

- Monitoring training progress
- Identifying overfitting or underfitting
- Comparing performance of different models

(Insert plots from your SDP-TRAIN-5-plots notebook here)

2.2.8 EXPLAINABILITY USING GRAD-CAM

Deep learning models often lack interpretability. To address this, **Grad-CAM (Gradient-weighted Class Activation Mapping)** is used.

Grad-CAM generates heatmaps that highlight the important regions in the X-ray image responsible for model predictions.

ADVANTAGES

- Improves transparency of the model
- Helps verify correctness of predictions
- Provides visual explanation for medical diagnosis
- Builds trust in AI-based systems

2.2.9 PREDICTION MECHANISM

During inference:

- The input X-ray image is passed through the trained model
- Probability scores are generated for each class
- Thresholding is applied to determine disease presence
- Grad-CAM visualization is generated

The final prediction includes multiple disease labels along with visual explanations.

2.3 STANDARDS

The proposed system is developed by following standard practices in deep learning and medical image analysis. The implementation is carried out using the Python programming language due to its simplicity and wide support for machine learning applications.

The system is built using the PyTorch framework, which provides efficient tensor operations, dynamic computation graphs, and GPU acceleration. These features help in training deep learning models effectively on large-scale image datasets. Supporting libraries such as NumPy, OpenCV, and PIL are used for image processing and data handling, while Matplotlib is used for visualization of results.

A. DATA PROCESSING STANDARDS

All input chest X-ray images are processed using standard preprocessing techniques. The images are resized to **224 × 224 pixels** to match the input requirements of pre-trained CNN models. Pixel values are normalized to ensure stable training and faster convergence. These preprocessing steps are essential to maintain consistency across the dataset and improve model performance.

B. MODEL TRAINING STANDARDS

The system follows standard deep learning training practices. Transfer learning is applied using pre-trained CNN models such as VGGNet, MobileNet, DenseNet, and EfficientNet-B0.

These models are trained on large-scale datasets like ImageNet and are fine-tuned for the chest X-ray classification task.

The training process includes:

- Forward propagation of input images
- Loss computation using Binary Cross Entropy
- Backpropagation for weight updates
- Optimization using the Adam optimizer

Regularization techniques such as **dropout and batch normalization** are used to reduce overfitting and improve generalization.

C. CLASSIFICATION STANDARDS

The problem is formulated as a **multi-label classification task**, where each image can belong to multiple disease classes. A sigmoid activation function is used in the output layer to generate independent probability scores for each class.

A threshold value is applied to convert probabilities into final predictions. This ensures that multiple diseases can be detected simultaneously from a single image.

The use of sigmoid activation allows each class to be treated independently, which is suitable for multi-label problems. A customized thresholding strategy is applied to improve sensitivity in detecting diseases. This approach helps reduce missed detections while maintaining reliable predictions.

D. EVALUATION STANDARDS

The performance of the models is evaluated using standard evaluation metrics such as:

- Training Loss
- Validation Loss
- Validation Accuracy

These metrics help in analyzing model performance and ensuring that the model generalizes well to unseen data.

E. EXPLAINABILITY STANDARDS

To improve the interpretability of deep learning models, the proposed system incorporates **Gradient-weighted Class Activation Mapping (Grad-CAM)** as an explainability technique. Grad-CAM generates visual heatmaps that highlight the important regions in the chest X-ray image which contribute most to the model's prediction.

This approach helps in understanding whether the model is focusing on clinically relevant areas such as lung regions or abnormal patterns, rather than irrelevant background features. It also provides a visual justification for the predicted disease labels, which is essential in medical applications.

In addition to improving transparency, Grad-CAM assists in identifying potential model errors and biases. If the model focuses on incorrect regions, it indicates the need for further training or dataset improvement. This makes the system more reliable and trustworthy.

Furthermore, explainability plays a critical role in healthcare systems, where decisions must be interpretable by medical professionals. By providing visual explanations, the system supports better human-AI interaction and increases confidence in automated predictions.

Overall, the integration of Grad-CAM ensures that the proposed model is not only accurate but also interpretable, aligning with modern standards in explainable artificial intelligence (XAI) for medical image analysis.

2.4 SYSTEM ARCHITECTURE

The system architecture of the proposed model is designed to perform multi-label classification of chest X-ray images using pre-trained convolutional neural network (CNN) architectures. The architecture integrates data preprocessing, feature extraction, classification, and explainability mechanisms into a unified framework to ensure accurate and interpretable predictions. A schematic representation of the system architecture is shown in **Figure 2.1**, while the internal CNN architecture is illustrated in **Figure 2.2**.

The input to the system consists of chest X-ray images, which are first processed through a preprocessing stage to ensure uniformity and compatibility with the CNN models. The preprocessing operations include resizing the images to a fixed resolution of 224×224

pixels, normalization of pixel intensities, and conversion into tensor representations. These steps are essential to standardize the input data and facilitate efficient learning.

The core component of the architecture is the feature extraction module, which utilizes pre-trained CNN models such as VGGNet, MobileNet, DenseNet, and EfficientNet-B0. These models serve as backbone networks that learn hierarchical feature representations from the input images. The initial convolutional layers capture low-level features such as edges and textures, while deeper layers learn high-level semantic features that are relevant for disease classification. The use of transfer learning allows the system to leverage pre-trained weights, thereby improving performance and reducing training time.

Following feature extraction, the high-dimensional feature maps are transformed using a global average pooling layer. This layer reduces the spatial dimensions of the feature maps while preserving the most salient information, thereby minimizing the risk of overfitting and reducing computational complexity. The pooled feature vectors are then passed to a fully connected layer, which acts as a classifier.

Since the problem is formulated as a multi-label classification task, the output layer employs a sigmoid activation function to generate independent probability scores for each disease class. This enables the model to predict multiple conditions simultaneously for a single input image.

To enhance interpretability, the architecture incorporates Gradient-weighted Class Activation Mapping (Grad-CAM) as an explainability module. Grad-CAM generates localization maps that highlight the regions of the input image contributing most significantly to the model's predictions. This provides valuable insights into the decision-making process of the model and ensures that the predictions are based on clinically relevant features.

Overall, the proposed system architecture provides a comprehensive and efficient framework for multi-label chest X-ray classification. By combining transfer learning, optimized feature extraction, and explainability techniques, the architecture achieves a balance between accuracy, efficiency, and interpretability, making it suitable for real-world medical applications.

2.5 WORKING FLOWCHART

The working flowchart of the proposed system illustrates the sequence of operations involved in performing multi-label chest X-ray disease classification. It provides a clear representation of how input data is processed through various stages of the system to generate final predictions. The flow begins with data acquisition and proceeds through preprocessing, model inference, and visualization of results.

Initially, chest X-ray images are provided as input to the system. These images are collected from the dataset and may vary in size and quality. Therefore, they are first passed through a preprocessing stage, where they are resized to a standard resolution of 224×224 pixels and normalized to ensure consistency. The preprocessed images are then converted into tensor format, making them suitable for input to the deep learning model.

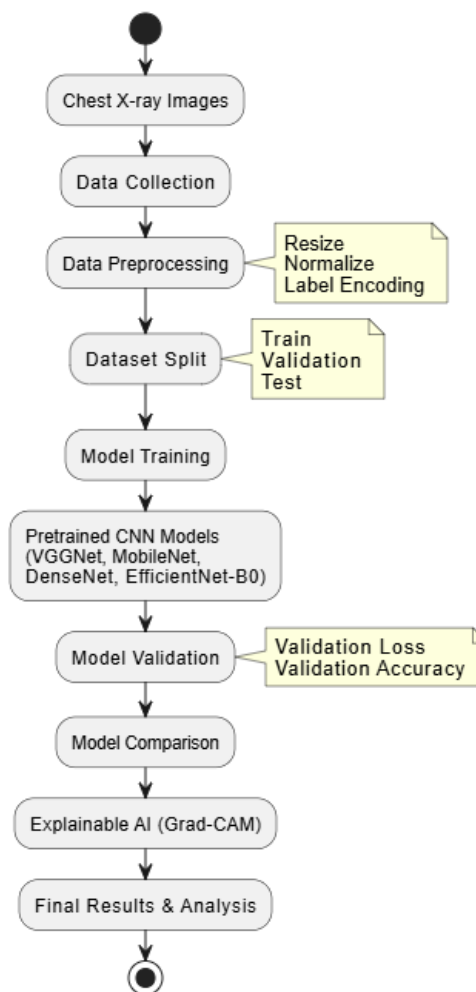


Figure 2.3: Working Flowchart

Following preprocessing, the images are fed into the trained convolutional neural network model. The model performs feature extraction using multiple convolutional layers, which capture both low-level and high-level patterns present in the X-ray images. These extracted features are then processed through subsequent layers, including pooling and fully connected layers, to generate prediction scores for each disease class.

Since the system is designed for multi-label classification, the output of the model consists of probability values corresponding to each class. A thresholding mechanism is applied to these probabilities to determine the presence or absence of specific diseases. This enables the system to identify multiple conditions within a single image.

To enhance interpretability, the system generates Grad-CAM visualizations for the predicted classes. These heatmaps highlight the regions of the X-ray image that contribute most to the model's predictions, providing insight into the decision-making process.

The final output of the system includes the predicted disease labels along with their corresponding probability scores and visualization maps. This flow ensures that the system not only provides accurate predictions but also offers transparency in its operation.

Overall, the working flowchart demonstrates the systematic progression of data through the model, from input acquisition to final prediction and visualization. It highlights the integration of preprocessing, deep learning inference, and explainability within a unified framework, ensuring efficient and reliable disease detection.

CHAPTER 3

IMPLEMENTATION

This chapter presents the implementation details of the proposed multi-label chest X-ray disease classification system. The implementation focuses on the practical realization of the model using deep learning frameworks, including dataset handling, preprocessing, model configuration, training setup, and evaluation procedures.

The system is implemented using the Python programming language due to its extensive support for machine learning and scientific computing. The deep learning models are developed using the PyTorch framework, which provides efficient tensor operations and supports GPU acceleration for faster training. Additional libraries such as NumPy, OpenCV, PIL, and Matplotlib are utilized for data handling, image processing, and visualization.

The dataset used for implementation is obtained from Kaggle, containing chest X-ray images labeled with multiple disease conditions. The dataset is organized in a structured format, where each image is associated with corresponding labels. During implementation, the labels are converted into a multi-label format, where each image is represented as a binary vector corresponding to the presence or absence of specific diseases. The dataset is divided into training and validation subsets to ensure proper evaluation of model performance.

Data preprocessing is implemented using transformation pipelines provided by PyTorch. All input images are resized to a fixed resolution of 224×224 pixels and normalized to ensure consistency across the dataset. The processed images are converted into tensor representations and loaded in mini-batches using data loaders, enabling efficient memory utilization and parallel processing during training.

The core implementation involves the use of pre-trained convolutional neural network architectures, including VGGNet, MobileNet, DenseNet, and EfficientNet-B0. These models are initialized with pre-trained weights and modified by replacing the final classification layer to match the required number of output classes. Transfer learning is applied by fine-tuning the model parameters, allowing the network to adapt to the chest X-ray dataset while retaining previously learned features.

The training process is implemented using an iterative optimization procedure. In each iteration, batches of input images are passed through the network to obtain predictions, and the loss is computed based on the difference between predicted and actual labels. The model parameters are updated using the Adam optimizer through backpropagation. The training is performed for a fixed number of epochs, and the model performance is monitored using training and validation metrics.

Model evaluation is carried out using validation data to assess the generalization capability of the trained models. Metrics such as training loss, validation loss, and validation accuracy are recorded during the training process. These metrics provide insights into model convergence and help identify potential issues such as overfitting or underfitting.

To further analyze model performance, visualization techniques are implemented. Graphs such as accuracy versus epochs and loss versus epochs are generated to illustrate the learning behavior of the models. These visualizations help in comparing different architectures and selecting the most effective model for the task.

In addition to quantitative evaluation, the system incorporates Grad-CAM for qualitative analysis. The implementation of Grad-CAM involves computing gradients of the final convolutional layer and generating heatmaps that highlight important regions in the input image. These visualizations provide insights into the model's decision-making process and ensure that predictions are based on relevant image features.

Finally, all components of the system are integrated into a unified pipeline for testing and inference. During inference, unseen chest X-ray images are processed through the trained model to generate predictions. The system outputs probability scores for each disease class along with corresponding Grad-CAM visualizations, enabling both accurate prediction and interpretability.

Overall, the implementation demonstrates the practical realization of the proposed system using deep learning techniques. The integration of transfer learning, efficient training procedures, and visualization mechanisms ensures that the system is both accurate and reliable for multi-label chest X-ray classification.

Model and Data Processing Code:

```
: train_tf = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.RandomHorizontalFlip(),
    transforms.ToTensor(),
    transforms.Normalize([0.485,0.456,0.406],
                          [0.229,0.224,0.225])
])

val_tf = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.ToTensor(),
    transforms.Normalize([0.485,0.456,0.406],
                          [0.229,0.224,0.225])
])

train_ds = NIHChestDataset(train_df, bbox_df, image_paths, train_tf)
val_ds   = NIHChestDataset(val_df, bbox_df, image_paths, val_tf)

train_loader = DataLoader(train_ds, batch_size=BATCH_SIZE, shuffle=True, num_workers=2)
val_loader   = DataLoader(val_ds, batch_size=BATCH_SIZE, shuffle=False, num_workers=2)

class EfficientNetXAI(nn.Module):
    def __init__(self):
        super().__init__()
        base = models.efficientnet_b0(weights="IMAGENET1K_V1")
        self.features = base.features
        self.pool = nn.AdaptiveAvgPool2d(1)
        self.classifier = nn.Linear(1280, len(CLASSES))

    def forward(self, x):
        feat = self.features(x)
        pooled = self.pool(feat).flatten(1)
        out = self.classifier(pooled)
        return out, feat

model = EfficientNetXAI().to(DEVICE)
```

Figure 3.1: Model and Data Processing Implementation using PyTorch

The implementation includes data transformation pipelines using PyTorch, where input images are resized, normalized, and augmented using techniques such as horizontal flipping. The dataset is loaded using custom dataset classes and `DataLoader` objects to enable batch processing and efficient training. The EfficientNet-B0 model is initialized with pre-trained weights, and its feature extraction layers are retained while modifying the final classification layer to match the number of disease classes. Adaptive average pooling is applied before the fully connected layer to reduce feature dimensions. During the forward pass, feature maps are extracted and passed through the classifier to obtain multi-label predictions. This modular implementation ensures flexibility and scalability for training different architectures. The implementation of the data preprocessing pipeline, data loaders, and EfficientNet-based model architecture is illustrated in Fig. 3.1.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the experimental results obtained from the implementation of the proposed multi-label chest X-ray disease classification system. A detailed analysis of the performance of different pre-trained convolutional neural network models is also provided.

The experiments are conducted using four pre-trained CNN architectures, namely VGGNet, MobileNet, DenseNet, and EfficientNet-B0. All models are trained using the same dataset, preprocessing techniques, and training parameters to ensure a fair comparison. The performance of these models is evaluated using training loss, validation loss, and validation accuracy.

During the training process, all models exhibit a gradual decrease in loss values along with an improvement in accuracy, indicating effective learning. The reduction in training and validation loss demonstrates that the models are able to capture meaningful patterns from the dataset. However, differences are observed among the models in terms of convergence speed and overall performance.

The performance of the models is summarized in Table 4.1.

Table 4.1: Model Performance Comparison

Model	Training Loss	Validation Loss	Validation Accuracy
VGGNet	0.2022	0.2091	0.8201
MobileNet	0.1446	0.1466	0.9424
DenseNet	0.1166	0.1372	0.5747
EfficientNet-B0	0.2260	0.2290	0.9091

From Table 4.1, it can be observed that **MobileNet achieves the highest validation accuracy** among all models. This indicates that MobileNet provides better generalization performance while maintaining computational efficiency. EfficientNet-B0 also shows strong performance by achieving a good balance between accuracy and model complexity. VGGNet, although effective, shows slower convergence due to its deeper architecture and

higher number of parameters. DenseNet demonstrates stable learning but achieves comparatively lower accuracy in this implementation.

To further analyze model performance, graphical visualizations are used. The accuracy versus epoch curve shows the improvement in model accuracy over training iterations. It is observed that MobileNet converges faster and reaches higher accuracy values compared to other models.

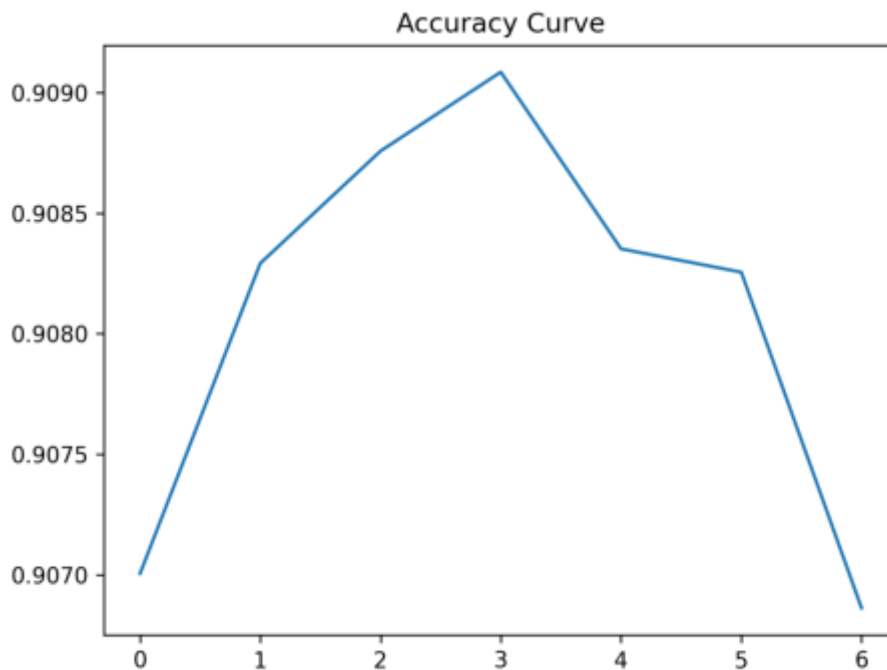


Figure 4.1: Accuracy vs Epoch graph

The accuracy versus epoch curve represents the variation in validation accuracy of the model over successive training epochs. As observed in Fig. 4.1, the accuracy initially increases, indicating that the model is effectively learning relevant features from the training data. The accuracy reaches its peak around the middle epochs, suggesting optimal learning performance at that stage. After this point, a slight fluctuation and minor decrease in accuracy can be observed, which may be due to overfitting or variations in validation data.

Overall, the curve demonstrates a stable learning trend, with accuracy values remaining relatively consistent after convergence. This indicates that the model has achieved a good balance between learning and generalization. The absence of large fluctuations confirms that the training process is stable and the model is not significantly underfitting or overfitting.

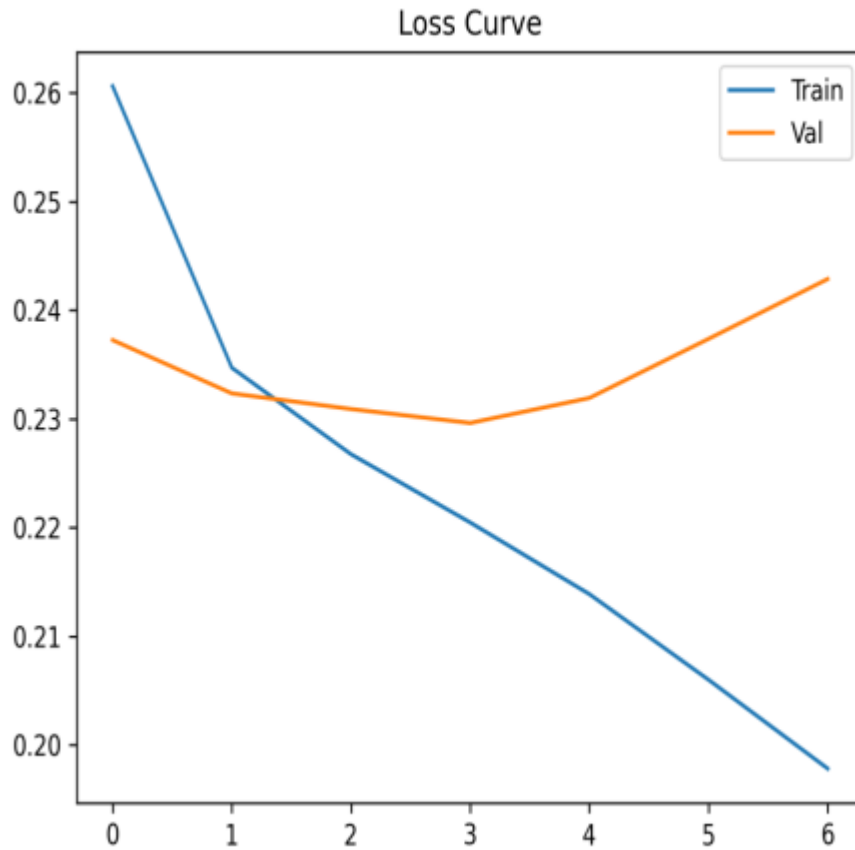


Figure 4.2: Loss vs Epoch graph

4.1 ADDITIONAL PERFORMANCE EVALUATION METRICS

A CLASS DISTRIBUTION

To understand the dataset characteristics, the distribution of different disease classes is analyzed. Medical datasets often suffer from class imbalance, where certain diseases have significantly more samples than others. This imbalance can affect model performance and bias predictions toward dominant classes.

The class distribution of the dataset is shown in Fig 4.3. It can be observed that some classes such as “No Finding” have a higher number of samples, while others like “Mass” and “Nodule” have relatively fewer samples.

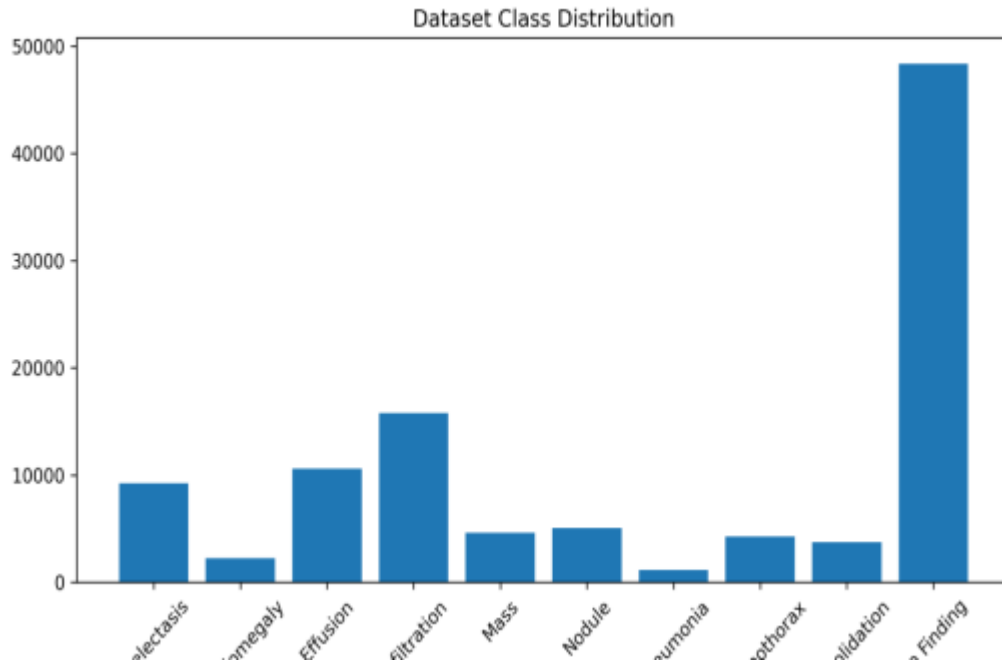


Figure 4.3: Dataset class distribution

B. CONFUSION MATRIX

To evaluate the classification performance in detail, confusion matrices are used for each disease class. The confusion matrix shows the number of correct and incorrect predictions made by the model, including true positives, true negatives, false positives, and false negatives.

The confusion matrices for selected classes are shown in Fig 4.4. These matrices indicate that the model is able to correctly classify most of the samples, while some misclassifications occur due to overlapping features between diseases.

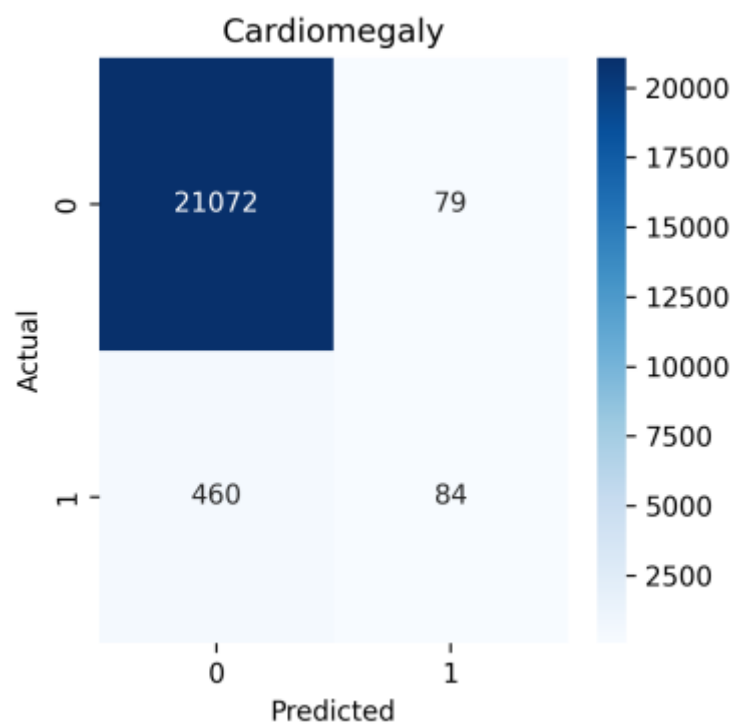


Figure 4.4: Confusion matrices for Cardiomegaly disease classes

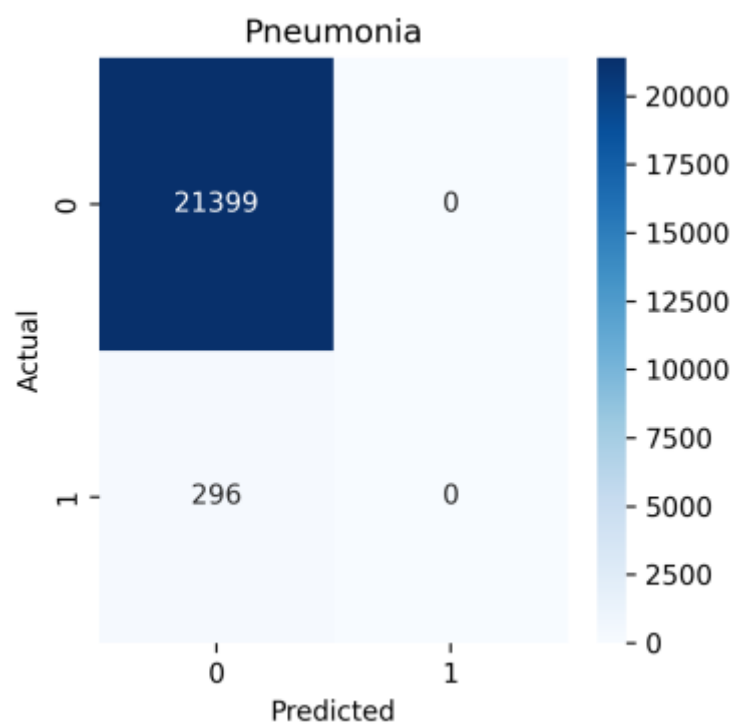


Figure 4.5: Confusion matrices for Pneumonia disease classes

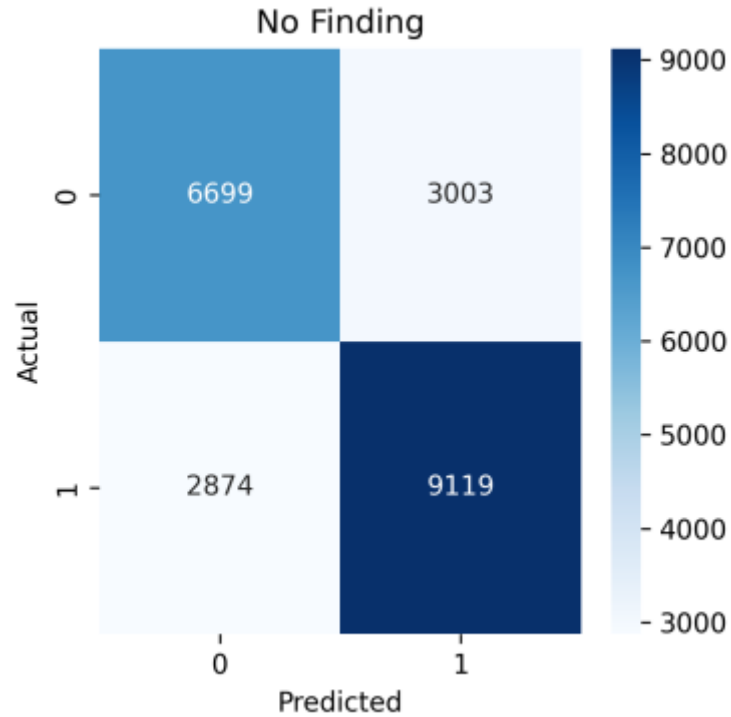


Figure 4.6: Confusion matrices for No Finding disease classes

C. ROC CURVE

Receiver Operating Characteristic (ROC) curves are used to evaluate the model's ability to distinguish between different classes in a multi-label classification setting. The ROC curve illustrates the relationship between the True Positive Rate (TPR), also known as sensitivity, and the False Positive Rate (FPR) at various threshold values. A higher TPR with a lower FPR indicates better model performance.

In this work, a combined ROC curve is generated for all disease classes to analyze the overall classification capability of the model. As shown in Fig 4.7, most of the ROC curves are positioned closer to the top-left corner of the plot, which indicates strong discriminative ability. This means that the model is capable of correctly identifying positive cases while minimizing false alarms.

The Area Under the Curve (AUC) is an important metric derived from the ROC curve, which summarizes the model's performance into a single value. A higher AUC value (closer to 1) indicates better classification performance. The obtained ROC characteristics suggest that the

model achieves high separability across multiple disease classes, demonstrating its effectiveness in handling complex medical image data.

Additionally, the ROC analysis is particularly useful in multi-label classification problems, where different threshold values can significantly impact prediction outcomes. By analyzing the ROC curves, it is evident that the model maintains a good balance between sensitivity and specificity, making it suitable for medical diagnosis applications where both false positives and false negatives are critical.

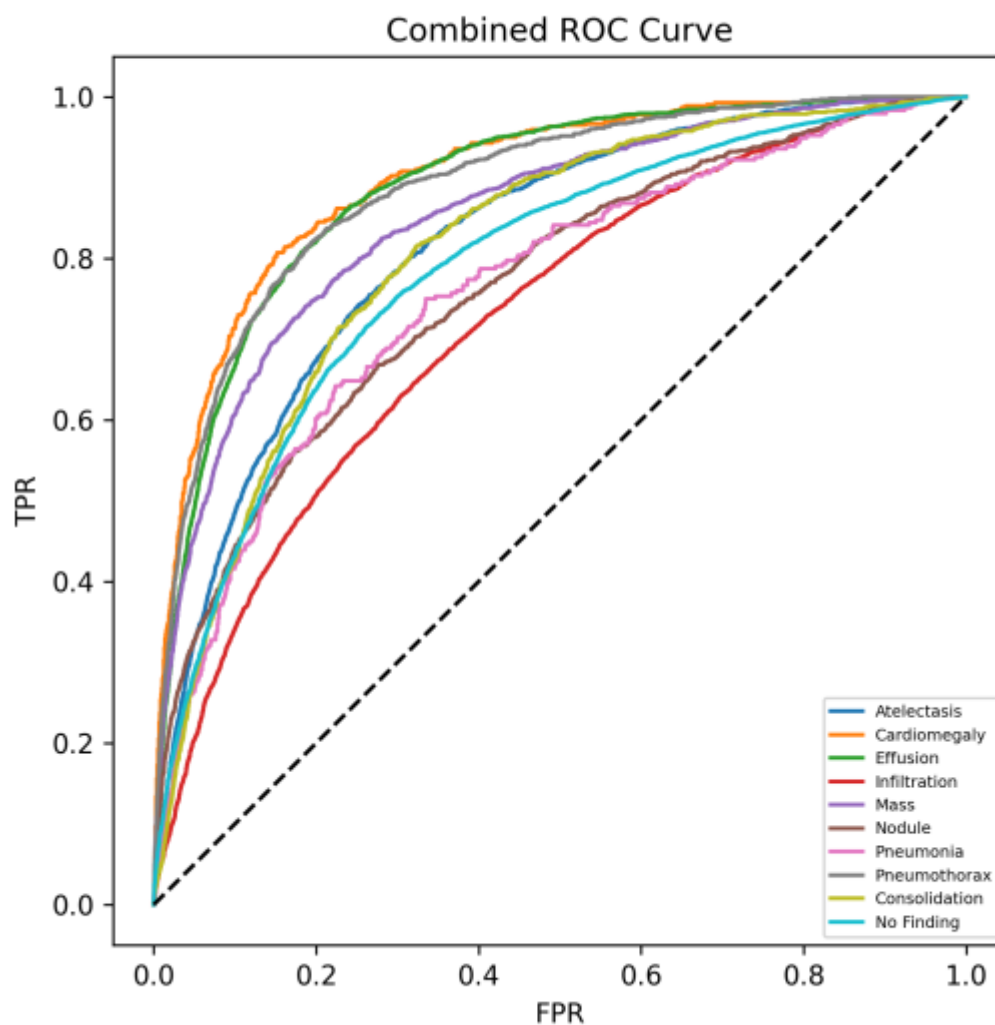


Figure 4.7: Combined ROC curve for all classes

D. PRECISION-RECALL CURVE

Precision-Recall (PR) curves are particularly useful for evaluating performance on imbalanced datasets. Precision indicates how many predicted positives are correct, while recall measures how many actual positives are correctly identified.

The Precision-Recall curves for selected classes are shown in Fig 4.8. These curves demonstrate that the model maintains a good balance between precision and recall, even in the presence of class imbalance.

A higher area under the Precision-Recall curve indicates better performance, especially for minority classes where correct detection is critical. The curves also show that the model is able to maintain reasonable precision at higher recall levels, which is important for reducing missed disease cases in medical diagnosis.

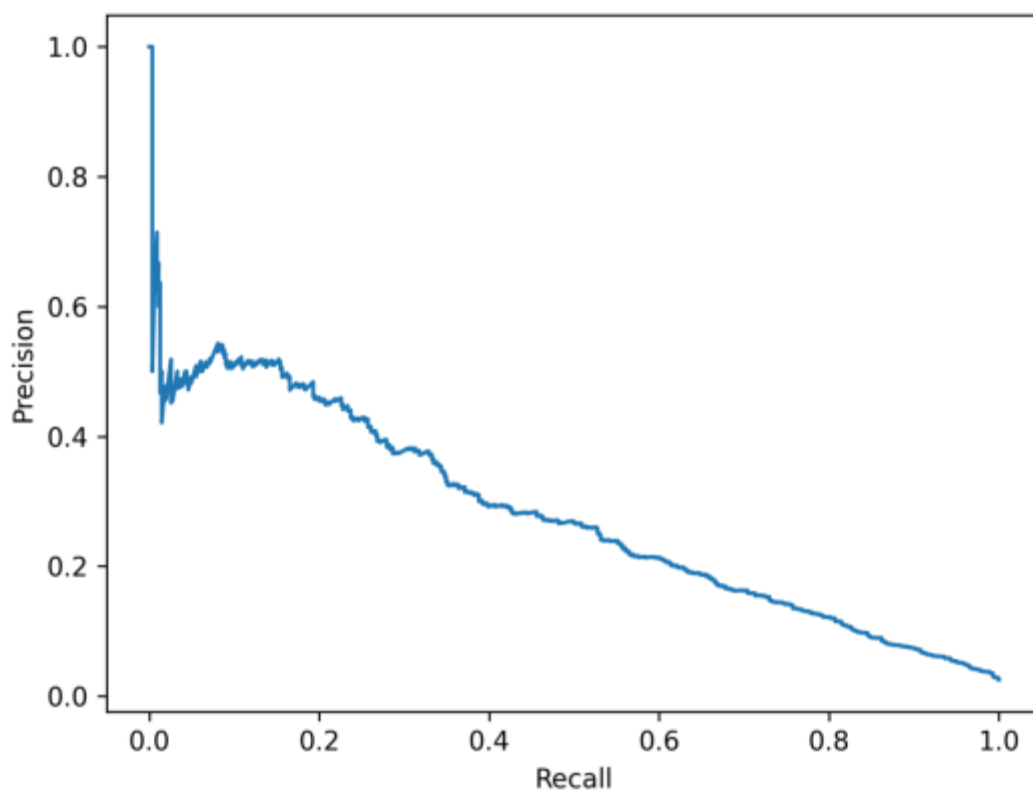


Figure 4.8: Precision-Recall curves for Cardiomegaly classes

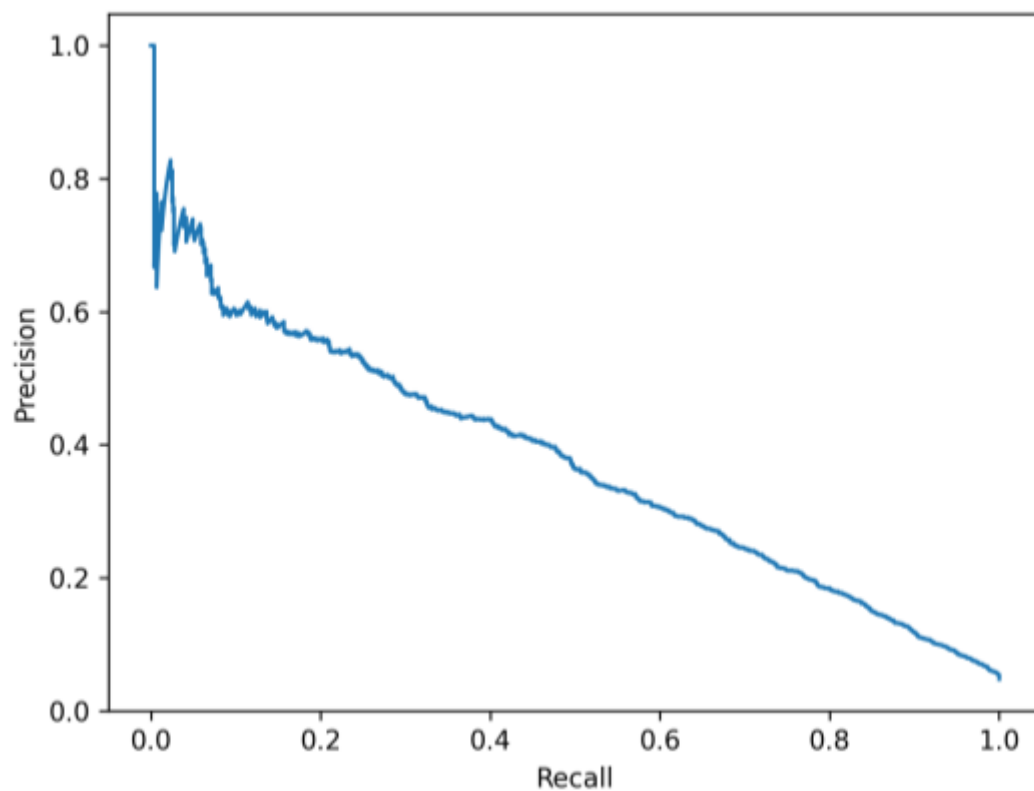


Figure 4.9: Precision-Recall curves for Pneumothorax classes

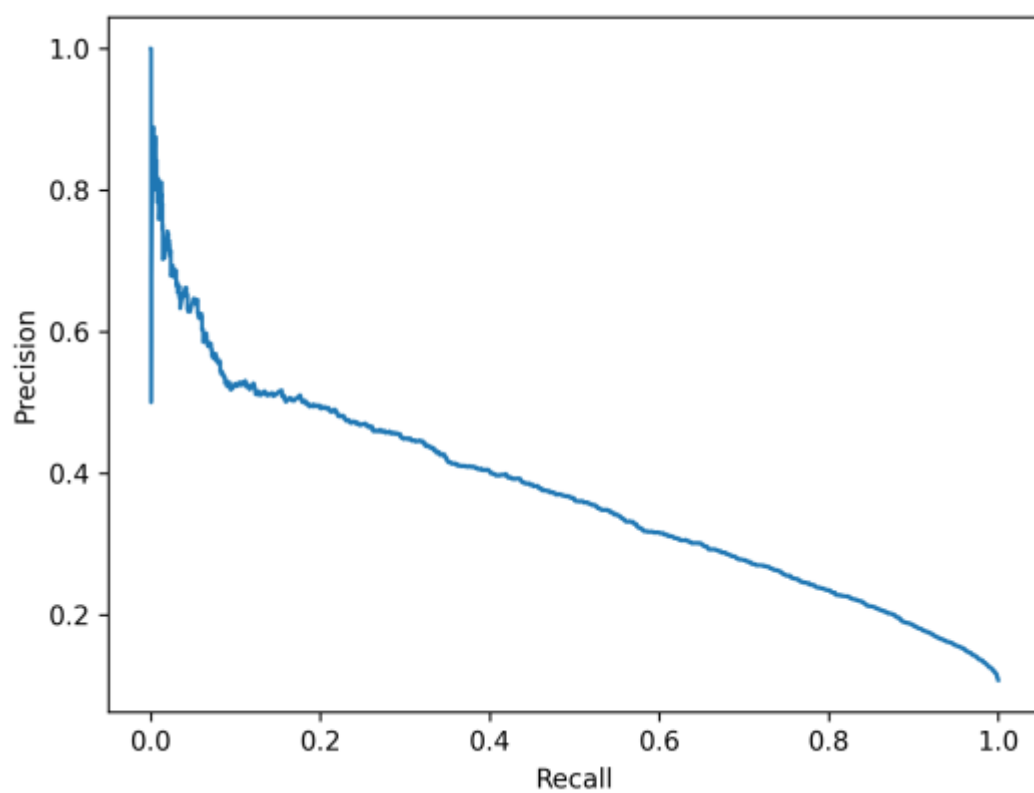


Figure 4.10: Precision-Recall curves for Atelectasis classes

E. ACCURACY PER CLASS

To further analyze model performance, class-wise accuracy is evaluated. This provides insight into how well the model performs on individual disease categories.

The class-wise accuracy is shown in Fig 4.11. It can be observed that the model performs better on certain classes compared to others, which may be influenced by dataset imbalance and feature similarity among diseases.

Classes with higher accuracy generally have a larger number of training samples, allowing the model to learn more representative features. In contrast, classes with lower accuracy may suffer from limited data availability or overlapping visual patterns with other diseases, making them harder to distinguish.

This variation in class-wise performance highlights the importance of balanced datasets and suggests that improvements can be achieved through techniques such as data augmentation, class weighting, or advanced architectures. Overall, the analysis confirms that the model performs reliably across most classes, there is scope for improvement in challenging categories.

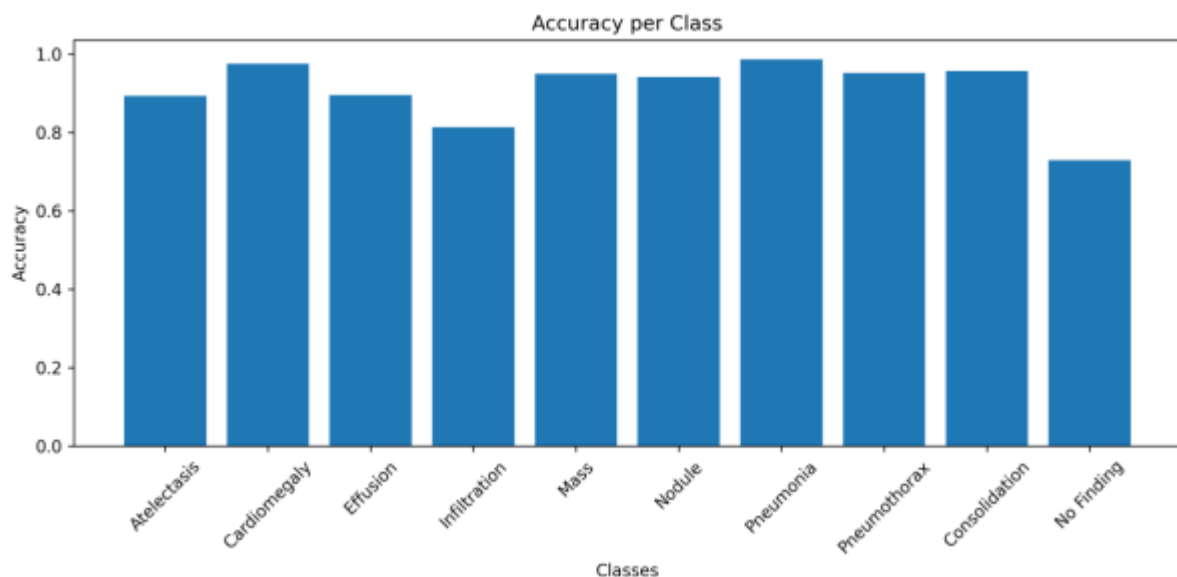


Figure 4.11: Accuracy per class

F. PREDICTION DISTRIBUTION

The distribution of predicted classes is also analyzed to understand the model's prediction behavior. This helps in identifying whether the model is biased toward specific classes.

The predicted class distribution is shown in Fig 4.12. The results indicate that the model predictions are reasonably distributed across different classes.

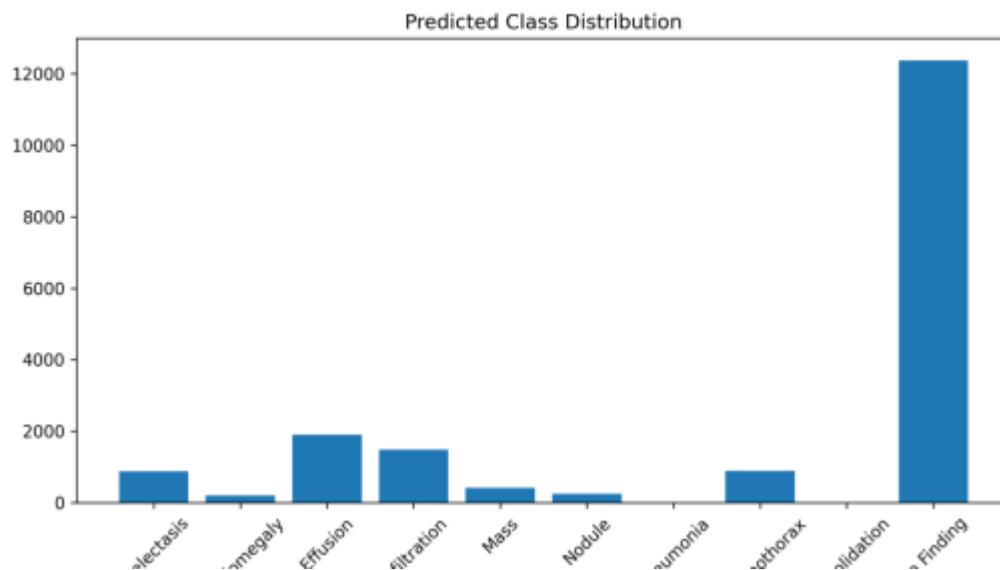


Figure 4.12: predicted class distribution

4.2 MODEL PREDICTION RESULTS

The following figures present sample prediction outputs generated by the proposed model, including probability scores for each disease class and the final classification result. For each input chest X-ray image, the model produces a probability value between 0 and 1 for all classes, indicating the likelihood of the presence of each disease.

Based on these probability scores, a thresholding mechanism is applied to determine the final prediction. The model first evaluates the “No Finding” class to identify normal cases. If the image is not classified as normal, the model then analyzes the remaining disease classes to detect possible abnormalities. This approach enables the system to handle multi-label classification effectively, where more than one disease can be present in a single image.

The displayed outputs also provide a clear understanding of how the model makes decisions by showing confidence levels for each class. Higher probability values indicate stronger confidence in the presence of a disease, while lower values suggest its absence. These prediction results demonstrate the model’s ability to identify both normal and abnormal conditions, as well as multiple co-existing diseases within the same chest X-ray image.



Figure 4.13: Predicted probabilities showing no disease detected (normal case)

The prediction results shown in Fig 4.14 illustrate a case where the model detects multiple possible diseases in a single chest X-ray image. The probability scores indicate a high likelihood of Cardiomegaly (0.7223) and a moderate probability of Effusion (0.3485), both exceeding the defined threshold values.

Since the “No Finding” probability is low (0.0909), the model proceeds to evaluate the remaining disease classes. Based on the applied thresholding strategy, the model identifies cardiomegaly as the primary condition and also suggests the possible presence of effusion. Other disease probabilities remain significantly lower, indicating weaker confidence for those classes.

This example demonstrates the effectiveness of the proposed multi-label classification approach, where the model can identify more than one disease from a single X-ray image. It also highlights the model's ability to provide confidence scores, enabling better interpretation and supporting clinical decision-making.

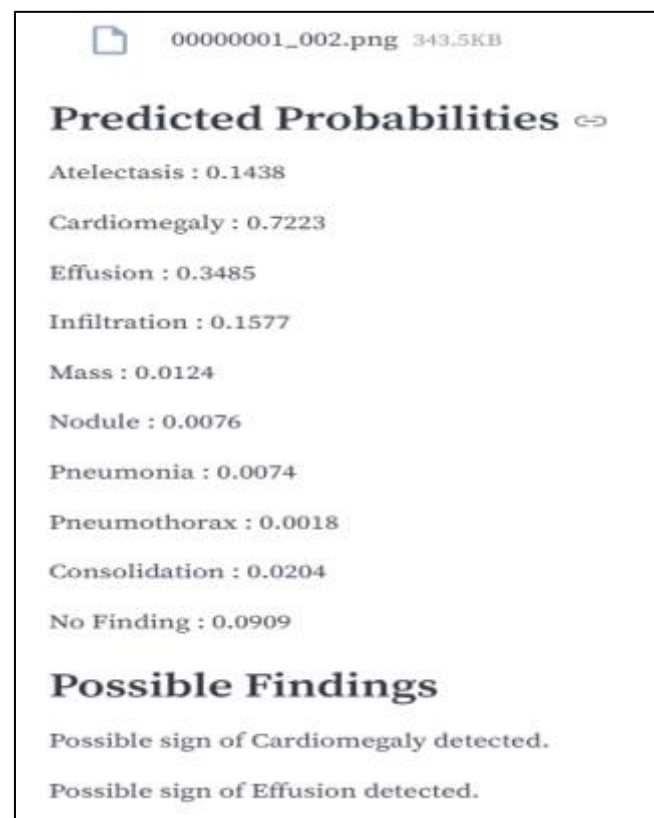


Figure 4.14: Predicted probabilities indicating possible cardiomegaly, effusion detection

The prediction results shown in Fig. 4.X represent a case where the model identifies multiple potential abnormalities in a chest X-ray image. The probability scores indicate a moderate likelihood of Infiltration (0.2998) and a high probability of Pneumothorax (0.5440), both exceeding the defined threshold values for disease detection.

Since the probability of “No Finding” is relatively low (0.2075), the model evaluates all disease classes. Based on the applied thresholding strategy, pneumothorax is identified as the primary condition, along with a possible indication of infiltration. Other disease probabilities remain comparatively low, indicating lower confidence in those classes.

This example demonstrates the capability of the proposed system to perform multi-label classification, accurately identifying multiple disease conditions from a single X-ray image. The results highlight the model's sensitivity in detecting critical conditions such as pneumothorax, which is essential for timely medical diagnosis.

Furthermore, the probability distribution indicates that the model assigns higher confidence to clinically significant abnormalities, thereby improving the reliability of predictions.



Figure 4.15: Predicted probabilities indicating possible infiltration, pneumothorax detection

The proposed system incorporates an input validation module to verify whether the uploaded image corresponds to a valid chest X-ray prior to processing, as shown in Fig. 4.6. This step ensures that only relevant medical images are forwarded to the classification model. By filtering out unrelated inputs, the system improves reliability and reduces the risk of

erroneous predictions. If a non-chest X-ray image is detected, the execution is terminated immediately. An appropriate error message is then displayed to inform the user and prevent further analysis.

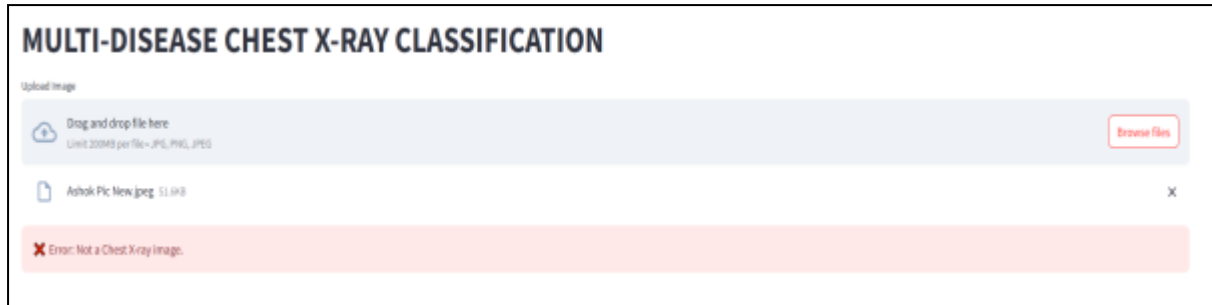


Figure 4.16: Error shown when a non-chest X-ray image is uploaded

A model comparison graph is used to visualize the performance differences among the evaluated models. It provides a clear comparison of their accuracy, helping identify the best-performing architecture.

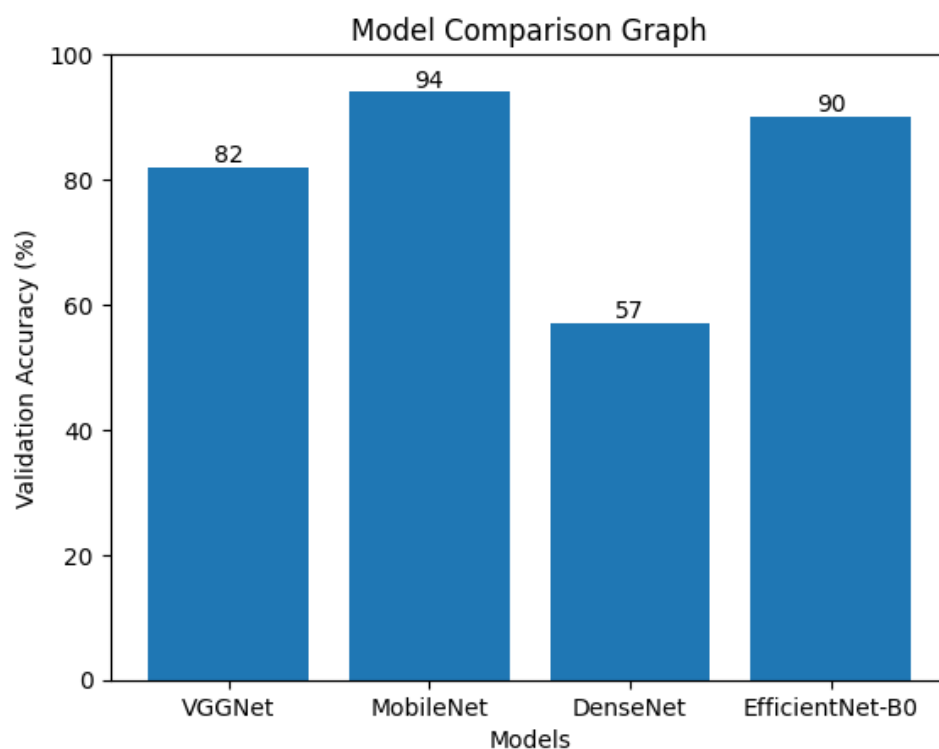


Figure 4.17: Model Comparison graph

In addition to quantitative evaluation, qualitative analysis is performed using Grad-CAM visualizations. The generated heatmaps highlight the regions of the chest X-ray images that contribute most to the model's predictions. It is observed that the model focuses on relevant lung regions and abnormal areas, indicating that it has learned meaningful and clinically relevant features for disease detection.

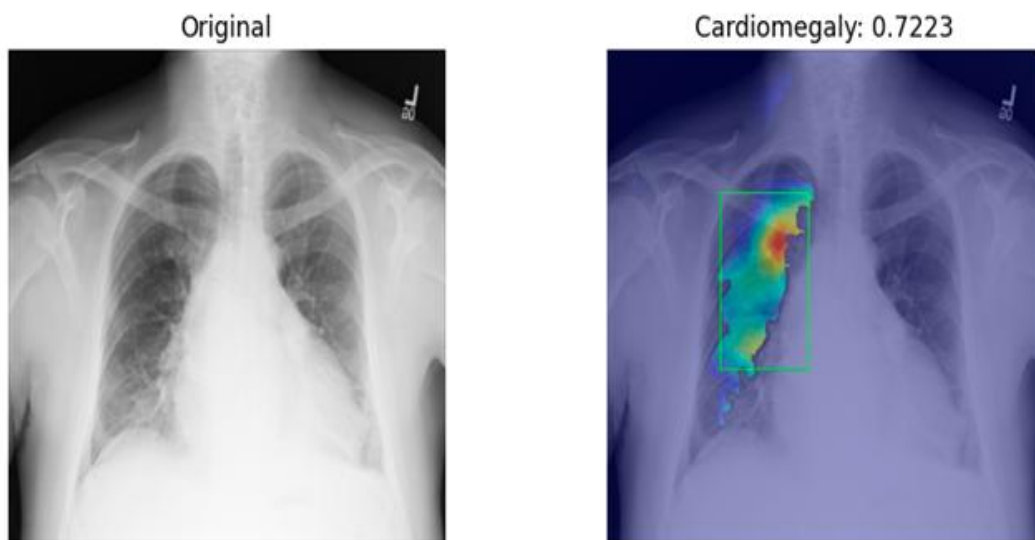


Figure 4.18: Grad-CAM results (Example 1)

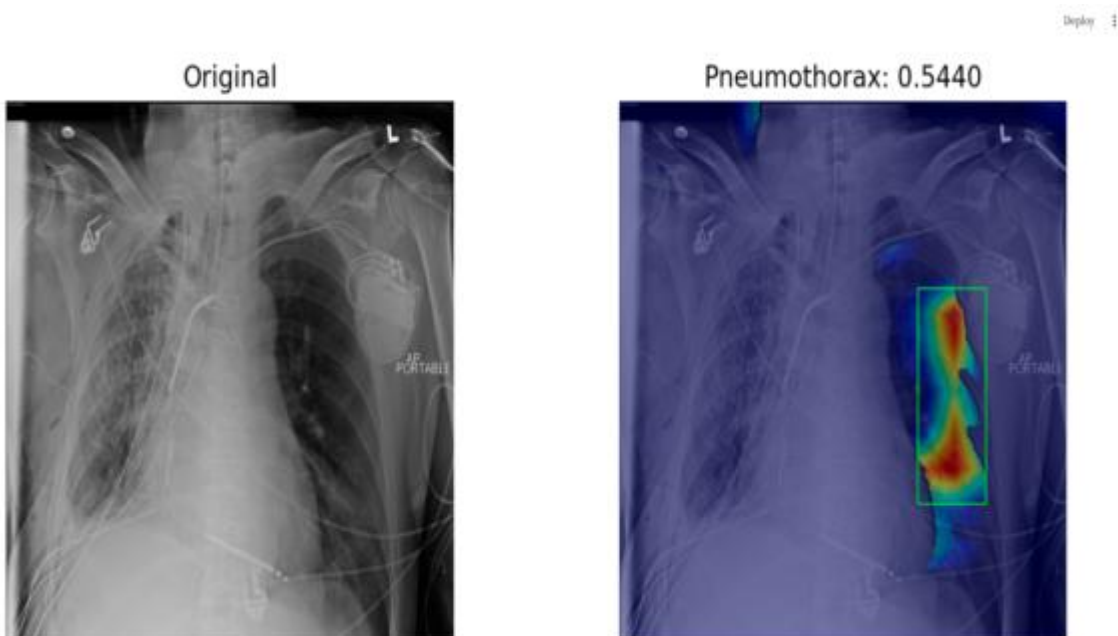


Figure 4.19: Grad-CAM results (Example 2)

The results show that transfer learning with pre-trained CNN models is effective for multi-label chest X-ray classification. Lightweight models like MobileNet achieve high performance with lower computational cost, while Grad-CAM improves interpretability.

Overall, the proposed system achieves reliable detection of multiple diseases from chest X-ray images. The combination of efficient model architectures, effective training strategies, and visualization techniques enhances the overall system performance. These results indicate that the proposed approach can serve as a supportive tool for automated medical diagnosis.

4.3 CHALLENGES FACED DURING EXPERIMENTS

The development and experimentation of the proposed multi-label chest X-ray classification system involved several practical and technical challenges. These challenges are categorized as follows:

1. DATASET-RELATED ISSUES

- The dataset obtained from Kaggle contains class imbalance, where some disease classes have significantly fewer samples compared to others
- Presence of noisy labels and weak annotations in medical datasets
- High similarity between different disease classes makes classification difficult
- Some images contain multiple overlapping conditions, increasing complexity in multi-label prediction

2. COMPUTATIONAL LIMITATIONS

- Limited access to high-performance GPUs resulted in longer training time
- Memory constraints during training large CNN models such as VGGNet and DenseNet
- Increased training time when experimenting with multiple models
- Resource limitations affecting hyperparameter tuning and experimentation

3. PREPROCESSING AND DATA HANDLING CHALLENGES

- Variations in image quality, contrast, and resolution across the dataset
- Need for proper normalization and resizing to maintain consistency
- Handling multi-label annotations and converting them into suitable format
- Efficient loading of large image datasets using data loaders

4. MODEL-RELATED CHALLENGES

- Difficulty in selecting the best-performing model among multiple architectures
- Some models showed slower convergence during training
- Risk of overfitting due to limited dataset size
- Balancing between model complexity and computational efficiency

5. MULTI-LABEL CLASSIFICATION CHALLENGES

- Predicting multiple diseases simultaneously increases complexity
- Choosing an appropriate threshold for classification
- Handling overlapping features between different diseases
- Difficulty in distinguishing subtle patterns in medical images

6. EXPLAINABILITY CHALLENGES (GRAD-CAM)

- Generating clear and meaningful heatmaps for all predictions
- In some cases, heatmaps highlight broader regions instead of specific abnormalities
- Interpreting Grad-CAM outputs for multiple labels simultaneously

Despite these challenges, the proposed system was successfully implemented and evaluated. The use of transfer learning, proper preprocessing, and model comparison helped in overcoming many of these issues and achieving reliable performance.

CHAPTER 5

CONCLUSION

The proposed project presents a comprehensive deep learning-based framework for multi-label classification of chest X-ray images using pre-trained Convolutional Neural Network (CNN) models. The primary objective of this work is to develop an efficient and reliable system capable of detecting multiple thoracic diseases from a single X-ray image, thereby assisting in automated medical diagnosis. The system leverages transfer learning techniques and evaluates multiple CNN architectures, including VGGNet, MobileNet, DenseNet, and EfficientNet-B0, to identify the most suitable model for this task.

The experimental results demonstrate that deep learning models are highly effective in extracting meaningful and hierarchical features from chest X-ray images. These models are capable of learning complex visual patterns such as textures, edges, and abnormal structures associated with different diseases. The ability of CNNs to automatically learn features eliminates the need for manual feature engineering and significantly improves classification performance.

Among the models evaluated, MobileNet achieves the best overall performance in terms of validation accuracy and convergence speed. Its lightweight architecture, based on depthwise separable convolutions, enables efficient computation while maintaining high predictive accuracy. This makes MobileNet particularly suitable for real-time and resource-constrained applications. EfficientNet-B0 also demonstrates strong performance by effectively balancing model depth, width, and resolution, thereby achieving good accuracy with optimized computational efficiency. In comparison, VGGNet, although effective in feature extraction, requires higher computational resources and shows slower convergence. DenseNet, while beneficial in feature reuse through dense connections, exhibits comparatively lower performance in this specific implementation.

The use of transfer learning plays a crucial role in enhancing model performance. By utilizing pre-trained weights from large-scale datasets such as ImageNet, the models are able to capture general image features and adapt them to the medical domain. This approach significantly reduces training time and improves generalization, especially when working with limited medical datasets. The preprocessing techniques applied, including image

resizing, normalization, and batch processing, further contribute to stable and efficient model training.

The system successfully addresses the challenge of multi-label classification, where a single chest X-ray image may contain multiple disease conditions. By using sigmoid activation in the output layer, the model is able to assign independent probability scores to each class, enabling simultaneous detection of multiple diseases. This reflects real-world clinical scenarios more accurately compared to single-label classification approaches.

In addition to quantitative performance evaluation, the integration of Grad-CAM enhances the interpretability of the system. Grad-CAM generates heatmaps that highlight the important regions of the chest X-ray images influencing the model's predictions. These visual explanations provide valuable insights into the decision-making process of the model and help verify that the model focuses on clinically relevant areas. This improves the transparency and trustworthiness of the system, which is essential in healthcare applications.

Despite achieving promising results, certain limitations are observed. The dataset used in this work contains class imbalance and variations in image quality, which can impact model performance. Some disease classes have fewer samples, making it difficult for the model to learn representative features. Additionally, computational limitations restrict extensive experimentation and fine-tuning of model parameters. The absence of additional evaluation metrics such as precision, recall, and F1-score limits the depth of performance analysis.

Overall, the proposed system demonstrates that deep learning-based approaches, particularly those utilizing transfer learning, can effectively perform multi-label classification of chest X-ray images. The combination of efficient CNN architectures, proper preprocessing techniques, and explainable AI methods results in a system that is both accurate and interpretable. This work highlights the potential of artificial intelligence in supporting medical professionals by providing faster and more consistent diagnostic assistance.

The developed system serves as a reliable and efficient solution for automated chest X-ray disease detection. It not only achieves strong classification performance but also provides meaningful insights through visualization techniques. The results of this study contribute to the growing field of AI-based medical imaging and demonstrate the feasibility of deploying deep learning models for real-world healthcare applications.

FUTURE WORK

Although the proposed system demonstrates effective performance in multi-label chest X-ray disease classification, several enhancements can be explored to further improve its accuracy, robustness, and real-world applicability.

One important direction is the use of larger and more diverse datasets. The current dataset suffers from class imbalance, which can negatively affect model performance, especially for rare disease classes. Incorporating balanced datasets and applying advanced data augmentation techniques can improve generalization.

Another potential improvement is the adoption of more advanced deep learning architectures and ensemble learning methods. Combining multiple models or utilizing transformer-based architectures can enhance feature extraction and classification performance.

The evaluation process can also be strengthened by incorporating additional performance metrics such as precision, recall, F1-score, and AUC-ROC. These metrics provide a more comprehensive understanding of model performance, particularly in multi-label classification scenarios.

Further research can focus on improving explainability by integrating more advanced Explainable AI (XAI) techniques beyond Grad-CAM. This can help provide deeper insights into model predictions and increase trust among medical professionals.

Finally, the system can be extended into a real-time application, such as a web-based or mobile-based platform, enabling practical deployment in clinical environments. Integration with hospital systems and validation using real-world clinical data can further enhance its usability and impact.

Overall, these future enhancements can significantly improve the reliability, efficiency, and applicability of the proposed system in real-world healthcare settings.

CHAPTER 6

APPENDIX

```
# =====  
# ❶ IMPORTS  
# =====  
  
import os, glob  
import numpy as np  
import pandas as pd  
from PIL import Image  
from tqdm import tqdm  
  
  
import torch  
import torch.nn as nn  
import torch.nn.functional as F  
from torch.utils.data import Dataset, DataLoader  
from torchvision import models, transforms  
  
  
from sklearn.model_selection import train_test_split  
from sklearn.metrics import (  
    multilabel_confusion_matrix,  
    classification_report,  
    roc_auc_score,  
    roc_curve,  
    precision_recall_curve  
)  
  
  
import matplotlib.pyplot as plt  
import seaborn as sns
```

```

torch.backends.cudnn.benchmark = True # 🚀 speed boost

# =====
# 🚀 CONFIG
# =====
DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
IMG_SIZE = 224
BATCH_SIZE = 32 # 🚀 increased (faster)
EPOCHS = 25
LR = 1e-4

DATA_ROOT = "/kaggle/input/datasets/organizations/nih-chest-xrays/data"

# =====
# 🚀 LOAD DATA
# =====
image_paths = {}
for folder in glob.glob(f"{DATA_ROOT}/images_*/images"):
    for img_path in glob.glob(f"{folder}/*.png"):
        image_paths[os.path.basename(img_path)] = img_path

df = pd.read_csv(f"{DATA_ROOT}/Data_Entry_2017.csv")
bbox_df = pd.read_csv(f"{DATA_ROOT}/BBBox_List_2017.csv")

CLASSES = [
    "Atelectasis", "Cardiomegaly", "Effusion", "Infiltration",
    "Mass", "Nodule", "Pneumonia", "Pneumothorax",
    "Consolidation", "No Finding"
]

```

```

for c in CLASSES:
    df[c] = df["Finding Labels"].apply(lambda x: 1 if c in x else 0)

df = df[df[CLASSES].sum(axis=1) > 0]
df = df[df["Image Index"].isin(image_paths.keys())]

train_df, val_df = train_test_split(df, test_size=0.2, random_state=42)

# =====
# 📁 DATASET
# =====

class NIH ChestDataset(Dataset):
    def __init__(self, dataframe, bbox_df, image_paths, transform):
        self.df = dataframe.reset_index(drop=True)
        self.bbox_df = bbox_df
        self.image_paths = image_paths
        self.transform = transform

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        row = self.df.loc[idx]
        img_path = self.image_paths[row["Image Index"]]

        image = Image.open(img_path).convert("RGB")
        image = self.transform(image)

        labels = torch.tensor(row[CLASSES].values.astype(np.float32))

```

```

bbox_mask = torch.zeros((IMG_SIZE, IMG_SIZE))
bboxes = self.bbox_df[self.bbox_df["Image Index"] == row["Image Index"]]

for _, b in bboxes.iterrows():
    x, y, w, h = int(b["Bbox x"]), int(b["y"]), int(b["w"]), int(b["h"])
    bbox_mask[y:y+h, x:x+w] = 1.0

return image, labels, bbox_mask.unsqueeze(0)

# =====
# 📦 TRANSFORMS
# =====
train_tf = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.RandomHorizontalFlip(),
    transforms.ToTensor(),
    transforms.Normalize([0.485,0.456,0.406],[0.229,0.224,0.225])
])

val_tf = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.ToTensor(),
    transforms.Normalize([0.485,0.456,0.406],[0.229,0.224,0.225])
])

train_loader = DataLoader(
    NIHChestDataset(train_df,bbox_df,image_paths,train_tf),
    batch_size=BATCH_SIZE,
    shuffle=True,

```

```

        num_workers=4,
        pin_memory=True
    )

val_loader = DataLoader(
    NIH ChestDataset(val_df, bbox_df, image_paths, val_tf),
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=4,
    pin_memory=True
)

# =====
# 📦 MODEL (WITH DROPOUT)
# =====

class EfficientNetXAI(nn.Module):
    def __init__(self):
        super().__init__()
        base = models.efficientnet_b0(weights="IMAGENET1K_V1")
        self.features = base.features
        self.pool = nn.AdaptiveAvgPool2d(1)

        self.classifier = nn.Sequential(
            nn.Dropout(0.3),
            nn.Linear(1280, len(CLASSES))
        )

    def forward(self, x):
        feat = self.features(x)
        pooled = self.pool(feat).flatten(1)
        out = self.classifier(pooled)

```

```

    return out, feat

model = EfficientNetXAI().to(DEVICE)

criterion = nn.BCEWithLogitsLoss()
optimizer = torch.optim.AdamW(model.parameters(), lr=LR, weight_decay=1e-4)

# =====
# 📖 HISTORY + EARLY STOPPING
# =====
train_losses, val_losses, val_accs = [], [], []

best_val_loss = float("inf")
patience = 3
counter = 0

# =====
# 📖 TRAINING
# =====
for epoch in range(EPOCHS):
    print(f"\n==== Epoch {epoch+1}/{EPOCHS} =====")

    model.train()
    train_loss = 0

    for imgs, labels, bbox in tqdm(train_loader):
        imgs, labels, bbox = imgs.to(DEVICE), labels.to(DEVICE), bbox.to(DEVICE)

    optimizer.zero_grad()

```

```

outputs, feats = model(imgs)

attn = feats.mean(dim=1, keepdim=True)
attn = F.interpolate(attn, size=(IMG_SIZE, IMG_SIZE), mode="bilinear")

loss = criterion(outputs, labels)
loss += 0.2 * F.mse_loss(attn, bbox)

loss.backward()
optimizer.step()

train_loss += loss.item()

# VALIDATION
model.eval()
val_loss, correct, total = 0, 0, 0

with torch.no_grad():
    for imgs, labels, _ in val_loader:
        imgs, labels = imgs.to(DEVICE), labels.to(DEVICE)
        outputs, _ = model(imgs)

        loss = criterion(outputs, labels)
        val_loss += loss.item()

    preds = torch.sigmoid(outputs) > 0.5
    correct += (preds == labels.bool()).sum().item()
    total += labels.numel()

```

```

val_acc = correct / total

train_losses.append(train_loss/len(train_loader))
val_losses.append(val_loss/len(val_loader))
val_accs.append(val_acc)

print(f"Train Loss: {train_losses[-1]:.4f}")
print(f"Val Loss : {val_losses[-1]:.4f}")
print(f"Val Acc : {val_acc:.4f}")

# 📌 EARLY STOPPING
if val_loss < best_val_loss:
    best_val_loss = val_loss
    counter = 0
    torch.save(model.state_dict(), "best_model.pth")
else:
    counter += 1
    if counter >= patience:
        print("Early stopping triggered")
        break

# =====
# 📌 PLOTS
# =====

os.makedirs("results", exist_ok=True)

plt.figure()
plt.plot(train_losses, label="Train")
plt.plot(val_losses, label="Val")
plt.legend()
plt.title("Loss Curve")

```

```

plt.savefig("results/loss.png", dpi=300)

plt.figure()
plt.plot(val_accs)
plt.title("Accuracy Curve")
plt.savefig("results/accuracy.png", dpi=300)

print("🎉 Training optimized + plots saved")

# =====
# 📁 CREATE RESULTS FOLDER
# =====

import os
os.makedirs("results", exist_ok=True)

# =====
# 📊 DATASET DISTRIBUTION
# =====

class_counts = train_df[CLASSES].sum()

plt.figure(figsize=(10,5))
plt.bar(CLASSES, class_counts)
plt.xticks(rotation=45)
plt.title("Dataset Class Distribution")
plt.savefig("results/class_distribution.png", dpi=300)

# Log scale version (paper-friendly)
plt.figure(figsize=(10,5))
plt.bar(CLASSES, class_counts)
plt.yscale("log")
plt.xticks(rotation=45)
plt.title("Class Distribution (Log Scale)")
plt.savefig("results/class_distribution_log.png", dpi=300)

# =====
# 📈 TRAINING CURVES
# =====

```

```

plt.figure()
plt.plot(train_losses, label="Train Loss")
plt.plot(val_losses, label="Val Loss")
plt.legend()
plt.title("Loss Curve")
plt.savefig("results/loss_curve.png", dpi=300)

plt.figure()
plt.plot(val_accs)
plt.title("Validation Accuracy")
plt.savefig("results/accuracy_curve.png", dpi=300)

# =====
# 📊 EVALUATION
# =====

model.eval()
all_labels, all_probs = [], []

with torch.no_grad():
    for imgs, labels, _ in val_loader:
        imgs = imgs.to(DEVICE)
        outputs, _ = model(imgs)
        probs = torch.sigmoid(outputs)

        all_labels.append(labels.cpu().numpy())
        all_probs.append(probs.cpu().numpy())

import numpy as np
y_true = np.vstack(all_labels)
y_probs = np.vstack(all_probs)
y_pred = (y_probs > 0.5).astype(int)

# =====
# 📊 METRICS
# =====

from sklearn.metrics import classification_report, roc_auc_score
print(classification_report(y_true, y_pred, target_names=CLASSES))
print("AUC:", roc_auc_score(y_true, y_probs, average='macro'))

```

```

# =====
# 📊 CONFUSION MATRICES
# =====

from sklearn.metrics import multilabel_confusion_matrix
import seaborn as sns

mcm = multilabel_confusion_matrix(y_true, y_pred)

for i, cm in enumerate(mcm):
    plt.figure()
    sns.heatmap(cm, annot=True, fmt="d")
    plt.title(f"CM - {CLASSES[i]}")
    plt.savefig(f"results/cm_{CLASSES[i]}.png", dpi=300)

# =====
# 📊 ROC CURVES (INDIVIDUAL)
# =====

from sklearn.metrics import roc_curve

for i in range(len(CLASSES)):
    fpr, tpr, _ = roc_curve(y_true[:, i], y_probs[:, i])
    plt.figure()
    plt.plot(fpr, tpr)
    plt.plot([0,1],[0,1], '--')
    plt.title(f"ROC - {CLASSES[i]}")
    plt.savefig(f"results/roc_{CLASSES[i]}.png", dpi=300)

# =====
# 📊 COMBINED ROC (IMPORTANT)
# =====

plt.figure(figsize=(6,6))

for i in range(len(CLASSES)):
    fpr, tpr, _ = roc_curve(y_true[:, i], y_probs[:, i])
    plt.plot(fpr, tpr, label=CLASSES[i])

plt.plot([0,1],[0,1], 'k--')
plt.xlabel("FPR")
plt.ylabel("TPR")

```

```

plt.title("Combined ROC Curve")
plt.legend(fontsize=6)
plt.savefig("results/combined_roc.png", dpi=300)

# =====
# 📊 PRECISION-RECALL CURVES
# =====
from sklearn.metrics import precision_recall_curve

for i in range(len(CLASSES)):
    precision, recall, _ = precision_recall_curve(y_true[:, i], y_probs[:, i])
    plt.figure()
    plt.plot(recall, precision)
    plt.title(f"PR - {CLASSES[i]}")
    plt.savefig(f"results/pr_{CLASSES[i]}.png", dpi=300)

# =====
# 📊 ACCURACY PER CLASS
# =====
accs = [(y_true[:, i] == y_pred[:, i]).mean() for i in range(len(CLASSES))]

plt.figure(figsize=(10,5))
plt.bar(CLASSES, accs)
plt.xticks(rotation=45)
plt.title("Accuracy per Class")
plt.savefig("results/accuracy_per_class.png", dpi=300)

# =====
# 📊 PREDICTION DISTRIBUTION
# =====
pred_counts = y_pred.sum(axis=0)

plt.figure(figsize=(10,5))
plt.bar(CLASSES, pred_counts)
plt.xticks(rotation=45)
plt.title("Predicted Class Distribution")
plt.savefig("results/pred_distribution.png", dpi=300)

print("📊 ALL POSSIBLE FIGURES GENERATED")

```

CHAPTER 7

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